FINITE-n CONVERGENCE AND PERIOD-4 OSCILLATORY STRUCTURE IN HOOK STATISTICS OF SELF-CONJUGATE PARTITIONS

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ABSTRACT. We present an experimental investigation of the finite-n convergence structure of the Craig–Ono–Singh asymptotic formula for t-hook statistics in self-conjugate integer partitions. Craig, Ono, and Singh proved that for fixed $t \geq 1$, the number of t-hooks among self-conjugate partitions of n is asymptotically normally distributed with explicit mean $\mu_t(n) \sim \frac{\sqrt{6n}}{\pi} - \frac{t}{2} + \frac{3}{\pi^2} + \frac{\delta_t}{4}$. While their work establishes large-n asymptotics, the finite-n error structure has not been computationally characterized. For $t = 1, 2, 3$ and $3 \leq n \leq 60$, we define the signed error $e(t, n) = \mu_t^{\text{emp}}(n) - \mu_t^{\text{theory}}(n)$. The scaled error $e(t, n)\sqrt{n}$ remains bounded, numerically verifying the theoretical $O(n^{-1/2})$ bound for the tested range. Furthermore, exploratory Fast Fourier transform analysis detects a frequency near 0.25 (period-4) in the SC error spectrum. This aligns with the expected analytic contributions from subdominant singularities at fourth roots of unity in the generating function. We propose a descriptive empirical ansatz for this period-4 oscillatory structure and outline the necessary statistical and analytic directions required to formalize it for larger n.

CONTENTS

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1. Introduction

The study of integer partitions sits at the crossroads of combinatorics, number theory, and mathematical physics. A partition of a positive integer n is a nonincreasing sequence of positive integers summing to n. Among the various combinatorial statistics attached to partitions, hook lengths occupy a distinguished role: they govern the representation theory of symmetric groups via the Frame–Thrall–Robinson hook length formula, appear in the Nekrasov–Okounkov identity, and connect to modular forms through the work of Han.

A partition λ is *self-conjugate* if its Young diagram equals its transpose. Self-conjugate partitions are in natural bijection with partitions into distinct odd parts via hook decomposition, a classical fact going back to Euler. Diagonal hooks in self-conjugate partitions are odd, though general hook lengths may be even or odd. This creates a structural constraint that distinguishes them sharply from unrestricted partitions.

Craig, Ono, and Singh [1] recently established that the number of t-hooks in self-conjugate partitions of n is asymptotically normally distributed, extending earlier work of Griffin, Ono, and Tsai on unrestricted partitions. Their proof uses two-variable generating functions derived via the Littlewood bijection for t-core partitions, together with Euler–Maclaurin summation, the dilogarithm function, and the saddle-point method.

The present work operates in a different regime. Rather than proving new asymptotic theorems, we conduct an exploratory computational investigation of the finite-n remainder term of the Craig–Ono–Singh formula. Specifically, we ask: how does the empirical mean $\mu_t^{\mathrm{emp}}(n)$ approach $\mu_t^{\mathrm{theory}}(n)$ as n grows?

The Craig–Ono–Singh asymptotic contains an uncharacterized remainder term of order $O(n^{-1/2})$. Consequently, the empirical residual studied here mixes finite-n fluctuations with the unknown higher-order asymptotic correction.

Our primary empirical observation is the numerical detection of a period-4 oscillatory component in the residual for the tested range $n \leq 60$. Rather than an unexpected anomaly, this structure is the anticipated analytic consequence of the arithmetic of self-conjugate Young diagrams, specifically arising from singularities at fourth roots of unity in the generating function.

Organization. Section 2 provides background on partitions, Young diagrams, hooks, and the Craig–Ono–Singh theorem. Section 3 describes our computational framework. Section 4 presents experimental observations: scaled error, FFT analysis, mod-4 decomposition, and model fitting. Section 5 compares SC and unrestricted partitions. Section 6 states the empirical model. Section 7 discusses methodological limitations and future directions.

2. Background

2.1. Partitions and Young diagrams.

A *partition* of n is a sequence $\lambda = (\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_k > 0)$ with $\sum_i \lambda_i = n$. We write $\lambda \vdash n$ and denote the number of partitions of n by $p(n)$.

The *Young diagram* of λ is the left-justified array of boxes with λ_i boxes in row i. The *conjugate partition* λ' is obtained by transposing this array.

2.2. Self-conjugate partitions.

A partition is *self-conjugate* if $\lambda = \lambda'$, i.e., the Young diagram is symmetric across its main diagonal. We denote by $\mathrm{SC}(n)$ the set of self-conjugate partitions of n and by $\mathrm{sc}(n) = |\mathrm{SC}(n)|$ their count.

Observation 1 (Euler, 1748)**.** $\mathrm{sc}(n)$ *equals the number of partitions of* n *into distinct odd parts.*

This is proved via hook decomposition: the diagonal hooks of a self-conjugate partition have odd lengths, and stripping them yields a bijection with distinct odd part partitions.

2.3. **Hook lengths.** For a box (i, j) in the Young diagram of λ, the *hook length* is

$$h(i, j) = (\lambda_i - j) + (\lambda'_j - i) + 1.$$

The *hook multiset* $H(\lambda)$ collects all hook lengths in λ.

For a fixed integer $t \geq 1$, define

$$N_t(\lambda) = \#\{h \in H(\lambda) : h = t\}.$$

In self-conjugate partitions, the diagonal hooks are odd; however, general hook lengths can be even or odd.

2.4. **The Craig–Ono–Singh theorem.** The following is the main result of [1].

Theorem 2 (Craig–Ono–Singh, 2025)**.** *Let $t \geq 1$ and let $\widehat{N}_{t,n}$ be the random variable giving the distribution of $N_t(\lambda)$ over $\mathrm{SC}(n)$. Then as $n \to \infty$, $\widehat{N}_{t,n}$ is asymptotically normally distributed with mean*

$$\mu_t(n) = \frac{\sqrt{6n}}{\pi} - \frac{t}{2} + \frac{3}{\pi^2} + \frac{\delta_t}{4} + O\left(\frac{1}{\sqrt{n}}\right)$$

and variance

$$\sigma_t(n)^2 = \frac{(\pi^2 - 6)\sqrt{6n}}{\pi^3} + O(1),$$

where $\delta_t = 1$ if t is odd and 0 otherwise.

3. Computational Framework

3.1. **Implementation.** All computations were performed in Python on standard hardware. The codebase includes:

- recursive partition generation,
- self-conjugacy testing via conjugate comparison,
- hook length computation for all boxes,
- empirical mean calculation per n and t,
- FFT analysis via `numpy.fft`,
- least-squares model fitting via `scipy.optimize.curve_fit`,
- entropy calculation for hook distributions.

3.2. **Definitions.** Fix $t \geq 1$. For each n, define the *empirical mean*

$$\mu_t^{\mathrm{emp}}(n) = \frac{1}{\mathrm{sc}(n)} \sum_{\lambda \in \mathrm{SC}(n)} N_t(\lambda).$$

Define the *signed error*

$$e(t, n) = \mu_t^{\mathrm{emp}}(n) - \mu_t^{\mathrm{theory}}(n),$$

where $\mu_t^{\mathrm{theory}}(n)$ is the leading asymptotic from Theorem 2.

For *unrestricted* partitions, the theoretical mean from Griffin–Ono–Tsai [3] is

$$\mu_t^{\mathrm{GOT}}(n) = \frac{\sqrt{6n}}{\pi} - \frac{t}{2} + \frac{3}{\pi^2} + O\left(\frac{1}{\sqrt{n}}\right).$$

Note the absence of the $\delta_t/4$ term compared to the self-conjugate case. We define the signed error for unrestricted partitions analogously using $\mu_t^{\mathrm{GOT}}(n)$.

Define the *relative error*

$$R(t, n) = \frac{|e(t, n)|}{|\mu_t^{\mathrm{theory}}(n)|}.$$

Define the *scaled error*

$$A(t, n) = e(t, n)\sqrt{n}.$$

3.3. **Computational range.** Experiments were run for $t \in \{1, 2, 3\}$ and $3 \leq n \leq 60$. For comparison, unrestricted partition hook statistics were computed over the same range.

Figure 1 illustrates the growth of $p(n)$, $p_{\text{odd}}(n)$, $p_{\text{distinct}}(n)$, and $\text{sc}(n)$ for $n \leq 20$, empirically confirming Euler's theorem ($p_{\text{odd}} = p_{\text{distinct}}$) and the self-conjugate–distinct odd parts bijection.

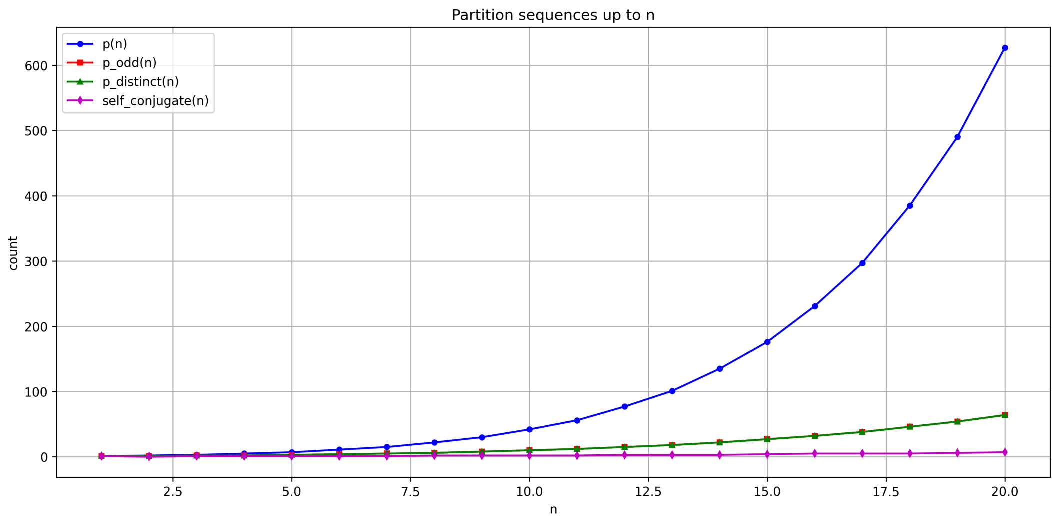

FIGURE 1. Growth of partition sequences. Red and green curves (p_odd and p_distinct) are visually identical, empirically confirming Euler's theorem.

4. Experimental Observations

4.1. **Euler's theorem and Young diagram bijection.** As a verification baseline, Figure 2 indicates that empirical mean hook length is consistently lower for self-conjugate partitions than for all partitions of n, consistent with the geometric compactness of self-conjugate Young diagrams.

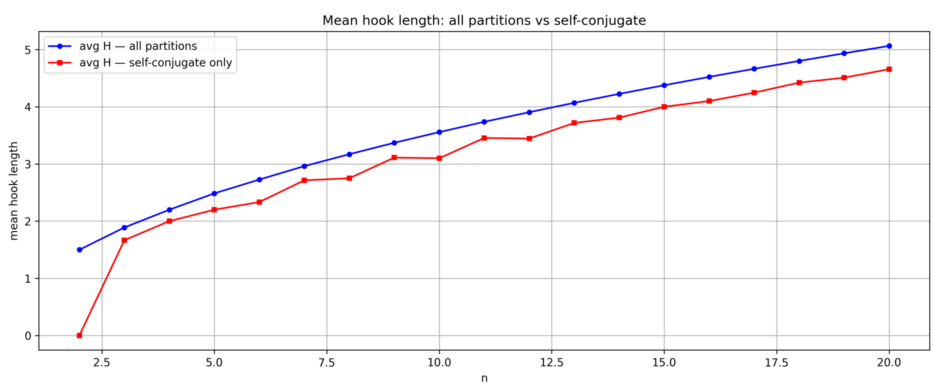

FIGURE 2. Mean hook length for all partitions vs. self-conjugate partitions. The gap is consistently negative, reflecting the geometric compactness of self-conjugate diagrams.

4.2. **Empirical vs. asymptotic mean.** Figure 3 illustrates the empirical mean $\mu_t^{\text{emp}}(n)$ alongside the Craig–Ono–Singh asymptotic $\mu_t^{\text{theory}}(n)$ for $t = 1, 2, 3$.

Figure 4 indicates the signed difference $e(t, n)$ for all three values of t.

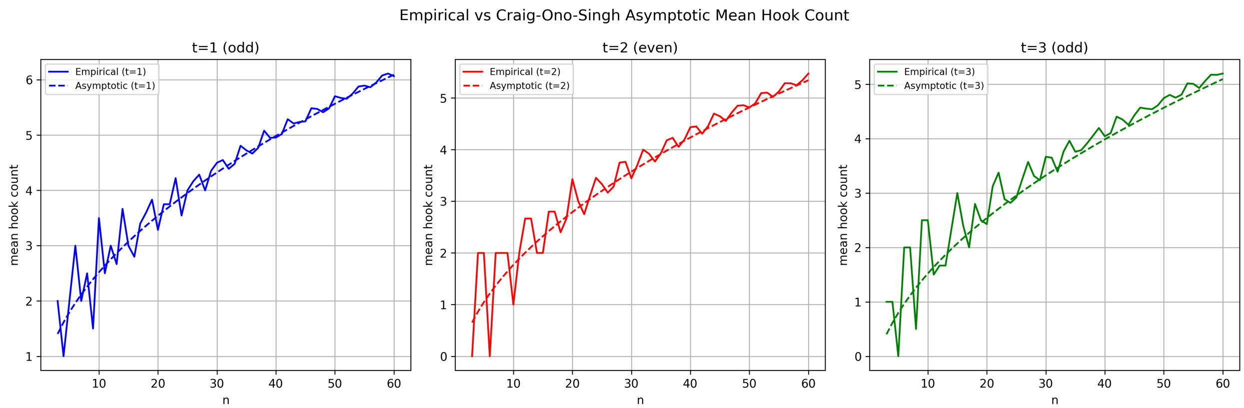

FIGURE 3. Empirical mean (solid) vs. Craig–Ono–Singh asymptotic (dashed) for $t = 1, 2, 3$. Convergence is visible but oscillatory.

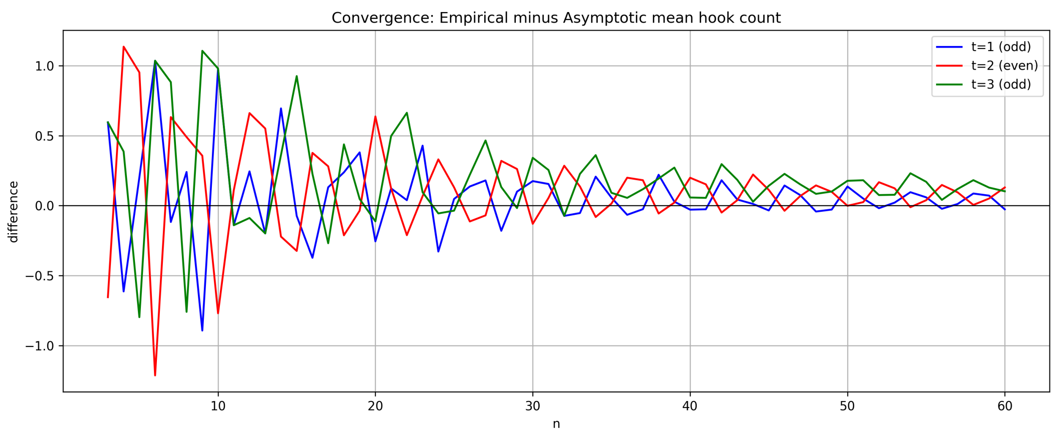

FIGURE 4. Signed error $e(t, n)$ for $t = 1, 2, 3$. The error oscillates around zero rather than decaying monotonically.

4.3. **Relative error decay.** Figure 5 illustrates $R(t, n)$ for $t = 1, 2, 3$. Even t exhibits a different decay profile than odd t. Because the $\delta_t/4$ constant is already subtracted from the theoretical mean, this difference confirms that the remaining higher-order asymptotic terms also depend strongly on the parity of t.

4.4. **Scaled error and decay rate.** Figure 6 indicates the scaled error $A(t, n) = e(t, n)\sqrt{n}$. The quantity remains bounded in the tested range, rather than decaying to zero or growing. This does not constitute a new bound, but successfully verifies the $O(n^{-1/2})$ decay rate established in Theorem 2 for the finite regime $n \leq 60$.

Figure 7 indicates $\log R(t, n)$. The roughly linear downward trend suggests polynomial rather than exponential decay.

4.5. **FFT analysis and period-4 structure.** Figure 8 illustrates the raw FFT of $e(t, n)$. A dominant peak appears near frequency 0.25, suggesting a period-4 oscillation. This peak appears to persist after parity averaging (Figure 9), ruling out pure period-2 (parity) as the sole source.

It must be noted that the raw signed error contains a non-stationary decaying trend. Applying an FFT to a short sequence of length 58 without detrending or tapering risks spectral leakage. The

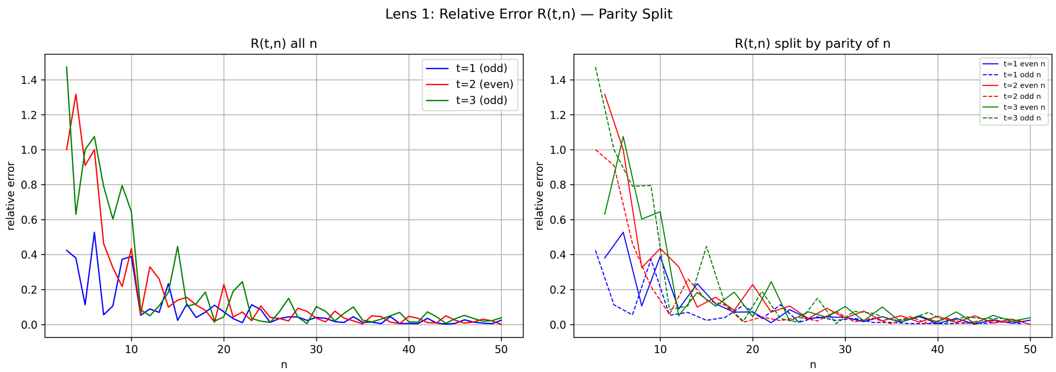

FIGURE 5. Relative error $R(t,n)$ and parity-of-n split. Odd t exhibits persistently higher relative error for $n < 30$.

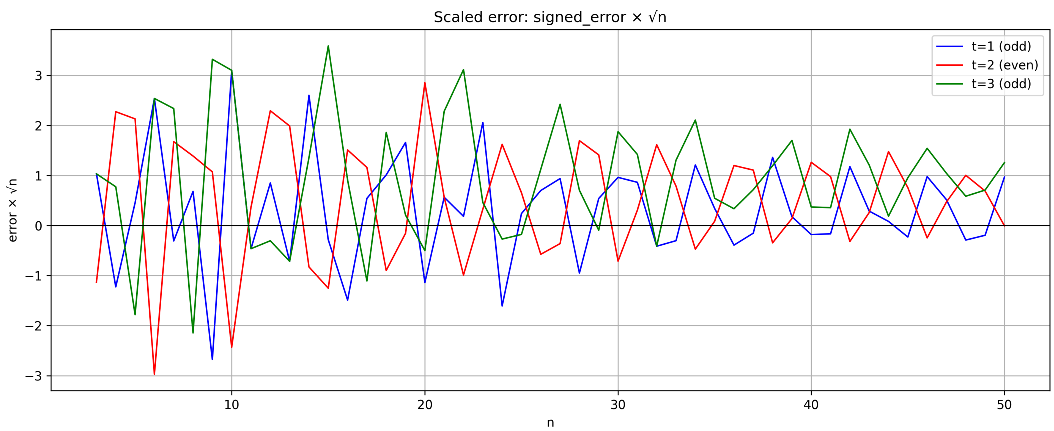

FIGURE 6. Scaled error $e(t,n)\sqrt{n}$ for $t = 1, 2, 3$. The quantity is bounded and oscillatory, numerically verifying the theoretical $O(n^{-1/2})$ decay.

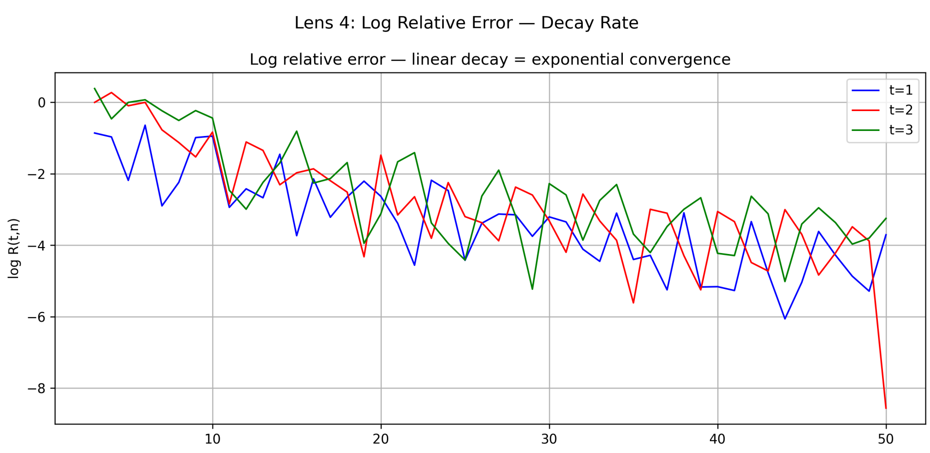

FIGURE 7. Log relative error. The downward linear trend suggests polynomial decay, not exponential.

peak at 0.25 is therefore an exploratory observation; future rigorous analysis on larger datasets must employ proper detrending to formally isolate the periodic signal.

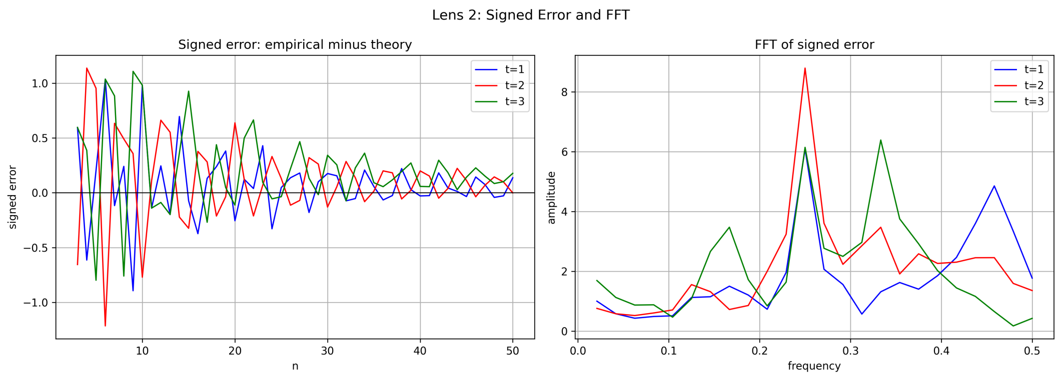

FIGURE 8. Left: signed error $e(t,n)$. Right: Raw FFT of $e(t,n)$. The dominant frequency is near 0.25 (period-4).

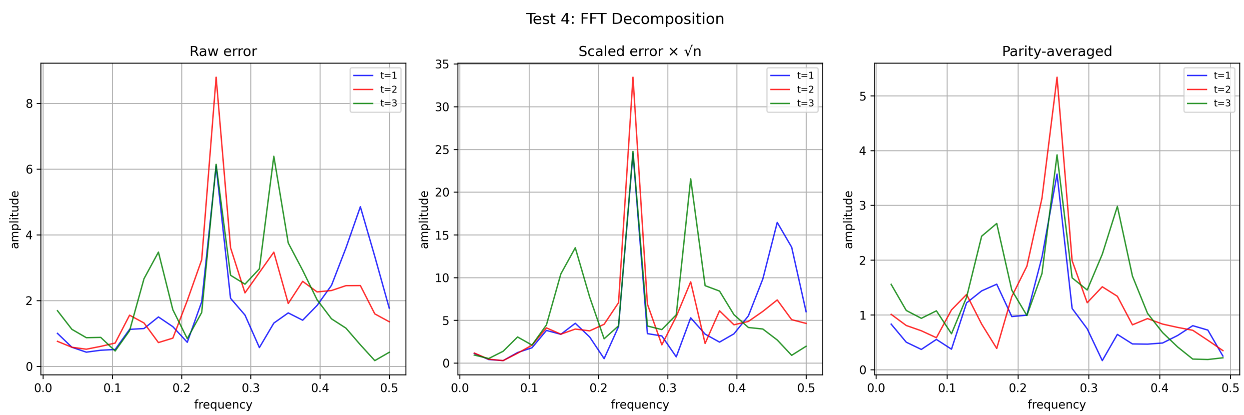

FIGURE 9. FFT of raw error, scaled error, and parity-averaged error. The period-4 peak at frequency 0.25 persists after parity averaging.

4.6. **Mod-4 residue analysis.** Figure 10 separates $e(t,n)$ by residue class of n modulo 4.

Table 1 indicates the mean of $A(t,n) = e(t,n)\sqrt{n}$ stratified by $n \bmod 4$, for both SC and unrestricted partitions.

4.7. **Oscillatory model fit.** We fit the model

$$e(t,n) \approx \frac{A_t + B_t \cos(\pi n) + C_t \cos(\pi n/2) + D_t \sin(\pi n/2)}{\sqrt{n}}$$

via least squares. Table 2 presents fitted constants and RMS residuals.
Note on model fitting. This is a saturated, overparameterized fit for a sequence of length 58. Without cross-validation or formal F-tests comparing it to a reduced trend-only model, the constants C_t, D_t risk overfitting. The model serves solely as a descriptive ansatz for $n \leq 60$.

Figure 11 illustrates the model fit overlaid on the empirical error.

FIGURE 10. Signed error separated by n mod 4. The four residue classes exhibit distinct behavior for SC partitions.

t	Type	$n \equiv 0$	$n \equiv 1$	$n \equiv 2$	$n \equiv 3$
1	SC	-0.440	-0.133	1.296	0.544
1	Unrestricted	1.810	1.859	1.826	1.847
2	SC	1.567	1.065	-0.775	0.105
2	Unrestricted	2.161	2.049	2.168	2.020
3	SC	0.125	0.516	1.862	1.269
3	Unrestricted	2.513	2.566	2.564	2.593

TABLE 1. Mean of $e(t,n)\sqrt{n}$ by residue class n mod 4. Given the small sample size per residue class (approx. 15 points), these means lack standard errors and formal ANOVA testing; they are presented as descriptive statistics rather than proven mod-4 effects.

t	Type	A_t	B_t	C_t	D_t	RMS
1	SC	0.390	0.245	-1.171	-0.416	0.213
1	Unrestricted	1.402	-0.030	-0.067	0.079	0.123
2	SC	0.400	-0.210	1.651	0.768	0.197
2	Unrestricted	1.628	0.168	-0.048	0.126	0.137
3	SC	0.895	0.175	-1.075	-0.556	0.277
3	Unrestricted	2.102	-0.058	-0.140	-0.031	0.134

TABLE 2. Fitted constants for the descriptive oscillatory model for $n \leq 60$.

5. Comparison: SC vs. Unrestricted Partitions

Figures 12–14 present the direct comparison between SC and unrestricted partitions.

The contrast in Table 1 is sharp, but requires careful interpretation. The unrestricted asymptotic mean used here is only a leading term; the actual error in the unrestricted case is known to be $O(1)$, not $O(n^{-1/2})$. The observed upward drift of the unrestricted scaled error (Figure 13) is exactly this $O(1)$ offset being multiplied by $\sqrt{n}$. Because this dominant non-oscillatory offset masks finer structure, the absence of a period-4 peak in unrestricted partitions should be interpreted cautiously.

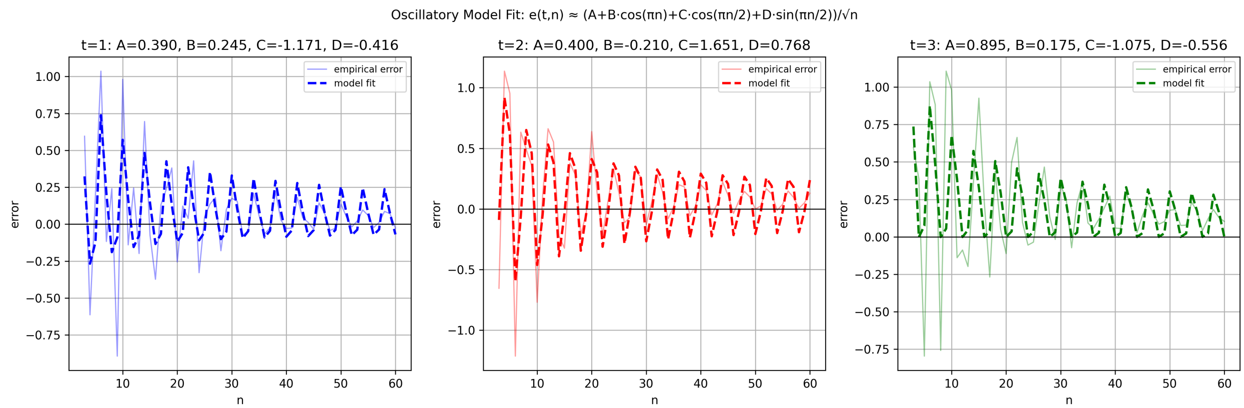

FIGURE 11. Empirical error (solid) and fitted oscillatory ansatz (dashed) for $t = 1, 2, 3$.

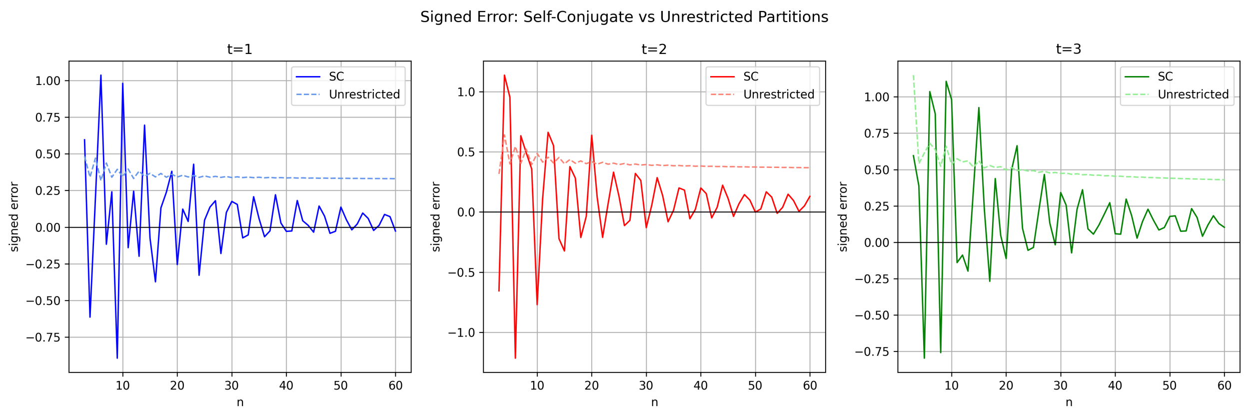

FIGURE 12. Signed error for SC (solid) vs. unrestricted (dashed) partitions.

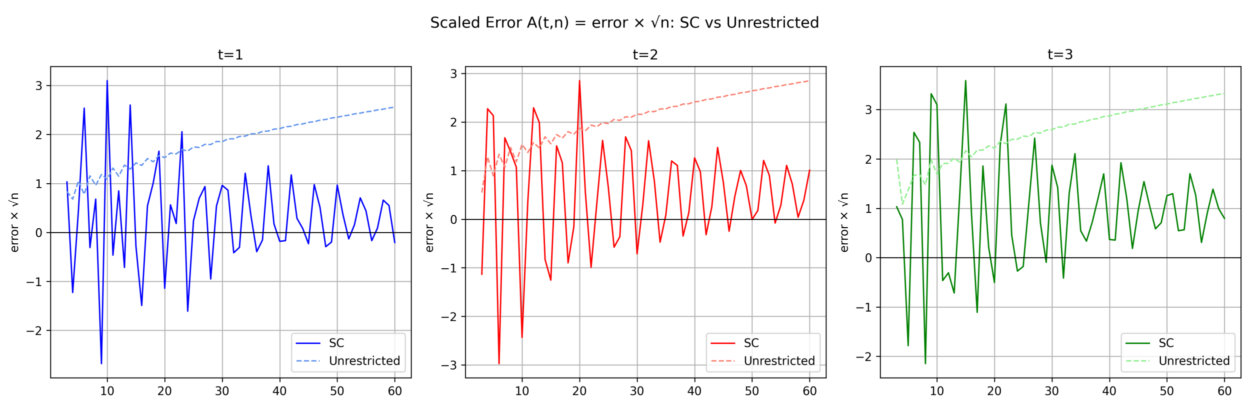

FIGURE 13. Scaled error $e(t, n)\sqrt{n}$ for SC vs. unrestricted. The unrestricted scaled error drifts upward.

To make the comparison fully meaningful in future work, the $O(1)$ constant must be extracted from the unrestricted asymptotic before raw periodicities are compared.

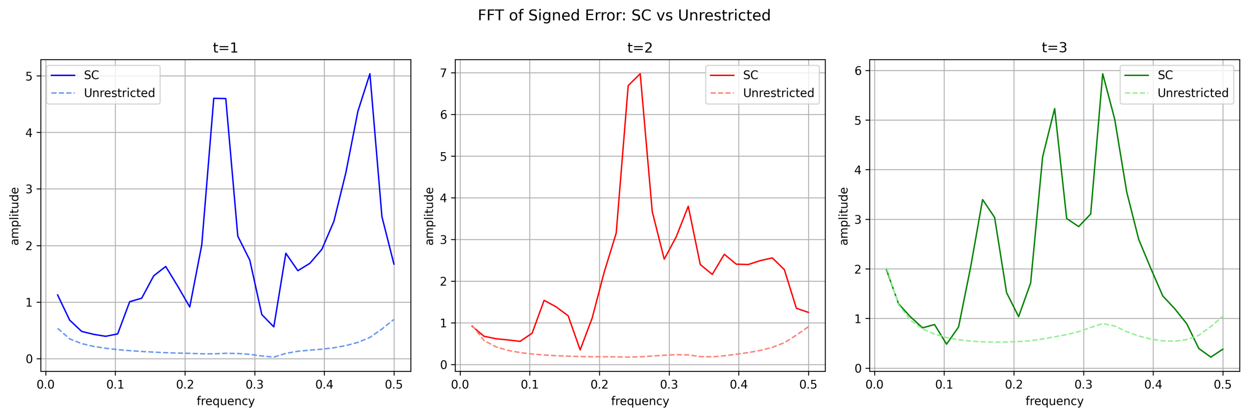

FIGURE 14. FFT comparison. SC partitions (solid) exhibit a peak near frequency 0.25. Unrestricted partitions (dashed) exhibit a nearly flat spectrum.

6. EMPIRICAL MODEL

The experimental evidence motivates the following empirical fit.

Observation 3. *Let $t \geq 1$ be fixed. Least-squares fitting suggests the descriptive empirical ansatz for the signed error $e(t,n) = \mu_t^{\text{emp}}(n) - \mu_t^{\text{theory}}(n)$ for self-conjugate partitions in the tested range $n \leq 60$:*

$$e(t,n)\sqrt{n} \approx A_t + B_t(-1)^n + C_t \cos\Big(\frac{\pi n}{2}\Big) + D_t \sin\Big(\frac{\pi n}{2}\Big)$$

for real constants A_t, B_t, C_t, D_t depending only on t.

Remark 4. *The period-4 terms $\cos(\pi n/2)$ and $\sin(\pi n/2)$ take the four values $(1, 0, -1, 0)$ and $(0, 1, 0, -1)$ on residue classes $n \equiv 0, 1, 2, 3 \pmod 4$ respectively, naturally producing the mod-4 amplitude structure observed in Table 1.*

Remark 5. *Rather than a purely unexpected discovery, the period-4 signal is an anticipated analytic feature. The generating function for self-conjugate partitions involves products of the form $(-q;q^2)_\infty$. When expanded near fourth roots of unity, the subdominant singularities at $q = \pm i$ naturally give rise to terms involving $\cos(\pi n/2)$ and $\sin(\pi n/2)$ via the circle method. Our numerical data serves to verify the presence of these expected subdominant contributions. The empirical model assumes constant amplitudes C_t, D_t in the scaled error, which serves as a descriptive ansatz for this finite range, though the true asymptotic amplitudes may exhibit slight n-dependence.*

7. DISCUSSION AND FUTURE DIRECTIONS

7.1. Summary of findings.

(1) The scaled error $e(t,n)\sqrt{n}$ remains bounded, numerically verifying the $O(n^{-1/2})$ error bound established by Craig–Ono–Singh for the tested range $n \leq 60$.
(2) Exploratory FFT analysis and mod-4 residue decomposition detect a period-4 oscillatory component in the finite-n residual.
(3) This period-4 structure aligns with the analytic contributions expected from subdominant singularities at fourth roots of unity in the generating function.
(4) Even t empirically observes a different decay profile than odd t, confirming that higher-order asymptotic terms depend strongly on parity.

7.2. **Limitations & Future directions.** These results are purely computational and exploratory. The range $n \leq 60$ is modest, and the statistical fitting risks overparameterization. To advance this exploratory note into rigorous asymptotic theory, the following steps are necessary:

- **Larger computational range.** Extend computations to $n \geq 10^3$ using optimized partition generation via the distinct odd parts bijection to verify if the periodic amplitudes stabilize.
- **Rigorous Statistical Testing.** Re-evaluate the FFT using proper detrending and windowing to avoid non-stationary spectral leakage, and apply formal ANOVA testing to the residue classes.
- **Theoretical derivation.** Extract the exact next term in the asymptotic expansion using the circle method near $q = \pm i$, and compare the predicted oscillatory coefficients directly with the empirical ones.
- **Refined Unrestricted Comparison.** Extract the $O(1)$ constant from the unrestricted asymptotic to allow for an unbiased comparison of periodic residuals between SC and unrestricted partitions.

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